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RESEARCH ARTICLE

A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment

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ABSTRACT Retrieval-Augmented Generation (RAG) pairs large language models with external search to constrain knowledge staleness and hallucination, a critical need in finance and e-commerce where numerical precision and regulatory auditability are non-negotiable. This study resembles four paradigms: sparse, dense, hybrid, and fusion retrieval, each realized as LangChain templates for cloud-hosted GPT-4o and on-premises LLaMA deployments. We embed a zero-shot baseline and assess every pipeline with the Retrieval-Augmented Generation Assessment (RAGAS) protocol, which scores faithfulness, relevancy, context quality, and latency without revealing proprietary gold data. Hybrid and fusion systems achieve the best combination of recall and faithfulness. Dense retrieval maximizes semantic coverage; sparse retrieval provides clearer, faster traceability. Each yields substantial gains over the baseline by anchoring profit metrics, compliance evidence, and pricing to audited documents. Although empirical evaluation primarily focuses on finance and e-commerce, the same blueprints and metrics can apply to healthcare, education, and legal technology. The domains in scope demand reasoning that is traceable, bias-aware, and up to date. This study, therefore, provides an auditable, production-ready framework for reliable RAG across knowledge-intensive settings.

INDEX TERMS Artificial intelligence, generative AI, large language models, retrieval-augmented generation.

I. INTRODUCTION

Large Language Models (LLMs) excel in zero- and few-shot generalization for tasks as translation and summarization [1]. However, their parameters may overlook recent large language models demonstrating strong zero- and few-shot generalization on tasks such as translation and summarization. Since parametric stores may be outdated and occasionally invent specifics, they pose risks in sensitive settings such as finance and e-commerce. Moreover, hallucinating specific data poses risks in sensitive areas such as finance and e-commerce. Retrieval-Augmented Generation (RAG) mitigates these risks by conditioning generation on passages retrieved at inference time, thus providing up-to-date con-

tent and auditable provenance [2]. Four retrieval paradigms dominate current practice. Sparse indices are transparent and fast but brittle to lexical variation [6]. Dense indexing projects text into a neural embedding space, improving semantic recall while reducing transparency and increasing compute cost [6]. Hybrid methods couple sparse and dense signals, trading added engineering for greater robustness [7]. Fusion pipelines chain retrieval stages to maximize coverage, at the expense of latency. Although these approaches are considerably studied, their comparative merits remain difficult to assess because prior evaluations (i) conflate retrieval and generation quality, (ii) under-specify key hyperparameters, and (iii) report aggregate scores without uncertainty quantification or systematic error analysis. The recently introduced RAGAS framework offers a principled, model-agnostic approach to separating retrieval quality from generation fidelity; however,

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published uses rarely include confidence intervals, paired significance testing, or robustness checks across retrieval loads and domains.

A. RESEARCH GAP

What is missing is a rigorous, reproducible comparison of these paradigms under realistic workloads that (a) pre-registers an evaluation protocol, (b) controls for tuning decisions that materially affect outcomes, (c) quantifies uncertainty and effect sizes, and (d) links engineering choices to measurable gains in faithfulness, recall, latency, and cost.

B. RESEARCH QUESTIONS

This study addresses the following:

- RQ1 (COMPARATIVE PERFORMANCE):
How do sparse, dense, hybrid, and fusion RAG pipelines compare on faithfulness and answer recall when generation is held constant?
- RQ2 (SENSITIVITY):
How sensitive are outcomes to retrieval depth, chunking strategy, and index tuning?
- RQ3 (LATENCY-UTILITY):
Under realistic service-level constraints, which paradigm delivers the best cost-adjusted utility for finance and e-commerce queries?
- RQ4 (AUDIBILITY AND PROVENANCE)
How do paradigms differ in citation coverage and source traceability (audibility) at the answer level?
- RQ5 (ROBUSTNESS):

Are observed advantages stable across two hosting modalities and across query difficulty strata?

C. CONTRIBUTIONS

The paper makes five contributions:

1) PROTOCOL

We introduce a pre-registered, task-agnostic evaluation harness that decouples retrieval from generation using RAGAS, reports bootstrap confidence intervals and paired significance tests, and includes power analysis to size experiments.

2) ABLATIONS AND SENSITIVITY

We provide controlled ablations over top-k chunk sizes, embedding families, and re-ranking, yielding Pareto frontiers that expose latency-quality trade-offs.

3) AUDIBILITY METRIC

We operationalize audibility as the fraction of answer claims supported by cited passages with confidence intervals, complementing faithfulness and answer recall.

4) IMPLEMENTATION TEMPLATES WITH INSTRUMENTATION

We release parameterized LangChain templates for each paradigm, featuring built-in telemetry for retrieval depth, latency, and cost, which have been validated on both cloud-

hosted GPT-4 and on-premises LLaMA-class deployments. Code utility is demonstrated by stress-testing templates under varied retrieval loads and task types, not merely by listing configurations.

5) DECISION SUPPORT VISUALS

We include comparative tables, bar/radar charts, and latency-utility curves, enabling practitioners to select pipelines under explicit budget and SLA constraints.

D. EVALUATION OVERVIEW

To make the protocol concrete, we (i) stratify queries by difficulty and domain (finance vs. e-commerce), (ii) balance answerable/unanswerable instances to prevent inflation of faithfulness via abstention, (iii) fix the generator within a study cell to isolate retrieval effects, (iv) report RAGAS faithfulness and answer recall with 95% bootstrap CIs, paired tests across paradigms, and error taxonomies and (v) provide latency and cost distributions per query. All index-level choices and dataset statistics are fully specified to ensure reproducibility.

E. SCOPE AND IMPACT

This study provides a clear plan for establishing reliable retrieval-augmented generation (RAG) pipelines in real-world settings. It combines architectural designs with a neutral evaluation method. The experiments focus on the finance and e-commerce sectors. However, this method can easily adapt to healthcare, education, and technology.

This study advances the evaluation of retrieval-augmented generation through a pre-registered and retrieval-isolating protocol that couples claim-level audibility with paired significance testing, bootstrap confidence intervals, and prospective power analysis under explicit service-level constraints. The design holds the generator fixed within each study cell, balances answerable and unanswerable items to curb abstention bias, and instruments retrieval templates with telemetry for hit positions, latency distributions, and token-cost traces. Unlike prior comparisons that report point estimates without uncertainty or that conflate retrieval and decoding quality, the present framework links concrete index choices and tuning knobs to measurable gains in faithfulness, recall, auditability, and latency across both cloud and on-premises deployments. The contribution is methodological, statistical, and deployment-oriented, rather than algorithmic, in the retriever itself.

II. LITERATURE REVIEW

Modern information-retrieval research spans early probabilistic models to neural ranking systems [10]. Within this lineage, Retrieval-Augmented Generation (RAG) emerged as a pivotal advance, explicitly coupling a large language model with an external retriever during inference. Lewis et al. inaugurated the paradigm by pairing a sequence-to-sequence generator with Dense Passage Retrieval (DPR), allowing access to millions of document fragments and resetting the

state of the art in open-domain question answering [2]. Subsequent studies refined negative sample strategies, unified retriever-generator training, and introduced memory-efficient vector stores, thus expanding the applicability of RAG to specialized domains, including finance and e-commerce [3].

A. BACKGROUND AND SCOPE

Modern information-retrieval (IR) has progressed from probabilistic and vector-space formulations to neural rankers and retrieval-conditioned generation, establishing a continuum from transparent lexical matching to semantic, acquired retrieval, and decoder-level fusion [10]. Within this scope, RAG formalized the practice of a generator with an external retriever at inference time, thereby refreshing knowledge and reducing hallucinations for knowledge-intensive tasks [2]. Subsequent work expanded the design space through better negatives and dual-encoder training for dense retrieval, large-scale memory stores, and late-interaction/reranking mechanisms that mediate the precision–recall trade-off in practical deployments [3], [4], [6], [7].

B. SPARSE RETRIEVAL

Sparse methods rank documents by lexical evidence, utilizing interpretable term-frequency, document-frequency, and normalization components, which yield latency on commodity hardware and auditable scores [5]. Their limitation is lexical structure, which is paraphrasing, synonymy, and cross-lingual variance, which may depress recall when query wording diverges from the corpus, particularly in regulated domains that value traceability and precise wording. Sparse methods often provide strong baselines yet benefit from complementary semantic retrieval to expand coverage.

C. DENSE RETRIEVAL

Dense Passage Retrieval (DPR) popularized dual encoders that match queries and contexts into an embedding space, enabling nearest-neighbor (ANN) search to provide evidence under considerable change [6]. This semantic reach improves recall for conversational, paraphrastic, and multilingual queries. However, it introduces higher indexing costs and opacity; similarity in the latent space is not easily decomposed into token-level justifications. These properties motivate hybridization with transparent signals and the use of rerankers to retain precision.

D. HYBRID AND LATE-INTERACTION APPROACHES

Hybrid pipelines combine lexical and dense signals, which compare contextualized query and passage token embeddings at inference time [4], [7]. Evidence from papers suggests that these combinations enhance effectiveness across long queries while incurring additional engineering and inference overhead [7]. In practice, hybrids are attractive when systems must preserve precision and auditability while expanding recall on semantically varied inputs.

E. FUSION AND DECODER-LEVEL CONDITIONING

Fusion systems generalize hybrid retrieval by aggregating multi-stage evidence either before decoding or within the generator (FiD), where the decoder attends over multiple retrieved passages [2], [9]. Query-expansion fusion probes different regions of the semantic space to mitigate under-specified queries, whereas FiD learns to weight evidence during the generation process. These designs typically yield the highest factual coverage and answer quality, albeit at the cost of latency and context-window pressure. They also complicate post-hoc auditing, as token-to-source attributions can be diffuse.

F. DOMAIN-DRIVEN DEPLOYMENTS

In finance and e-commerce, production RAG systems combine structured and unstructured sources to answer time-sensitive questions. Studies show reduced hallucinations when grounding is evaluated using protocolized metrics, such as RAGAS, if retrieval parameters are tuned to domain-specific idiosyncrasies (e.g., ticker/SKU tokens, multilingual traffic, freshness) [3], [5]. These domains thus serve as stress tests for precision, provenance, and latency under realistic service-level objectives.

G. IDENTIFIED GAPS AND POSITIONING

Despite steady progress, four methodological gaps recur in the literature and tooling that this paper addresses:

1) ENTANGLEMENT OF RETRIEVAL AND GENERATION

Many evaluations aggregate outcomes without isolating retrieval effects, which obscures the architectural choices that drive gains. We adopt a model-agnostic protocol that fixes the generator within each study cell and reports retrieval-conditioned metrics.

2) UNDER-SPECIFIED EXPERIMENTAL KNOBS

Key determinants of performance, top- k , chunking/overlap, BM25, encoder family, and dimensionality, ANN index type/parameters, reranking/fusion settings, are frequently omitted or only qualitatively described. We enumerate and ablate these factors to surface sensitivity and enable exact replication.

3) LIMITED UNCERTAINTY AND ERROR ANALYSIS

Prior reports often present point estimates without confidence intervals, paired significance tests, or error taxonomies. We incorporate bootstrap CIs, paired tests, and structured error breakdowns alongside an audibility measure.

4) INSUFFICIENT DEPLOYMENT-ORIENTED EVIDENCE

Code releases typically provide templates but little characterization of behavior under varied retrieval loads or latency constraints. We instrument templates with telemetry and present comparative tables and radar/latency plots to inform system selection under explicit SLAs.

Taken together, the prior art establishes the building blocks, sparse transparency, dense semantic reach, hybrid robustness, and fusion-stage synthesis, while leaving open a reproducible, statistics-aware comparison that clarifies how architectural choices trade off faithfulness, recall, latency, and auditability in high-stakes domains. The remainder of this paper fills that gap with an evaluation design that is both replicable and deployment-relevant, and that explicitly reports the methodological details reviewers have required.

III. MATERIALS AND METHODS

The zero-shot baseline utilizes the same generator as the RAG conditions, with a temperature of 0.2 and a maximum token limit of 512. The prompt template is identical to the RAG template except that the context block is empty. Assessment uses the same metrics, claim extraction procedure, and auditing rules as those applied to the RAG pipelines.

Retrieval-Augmented Generation (RAG) is a spectrum whose endpoints are strictly lexical retrieval and multistage, semantically rich fusion. At one extreme, sparse retrievers rely on inverted indices, such as TF-IDF or BM25, which excel at exact-term matching and offer fully traceable scoring [6]. On the other hand, fusion pipelines orchestrate several retrieval stages, query reformulation, cross-index aggregation, or decoder-level attention to assemble an evidence set that is simultaneously broad and precise. Between these poles lie dense hybrid strategies that trade storage overhead for semantic reach. The following subsections outline the end-to-end workflows, LangChain implementations, and qualitative advantages and disadvantages of each paradigm. A concluding table summarizes their relative performance under the Retrieval-Augmented Generation Assessment (RAGAS) protocol.

A. DATASET CONSTRUCTION

Finance and e-commerce form testbeds since the accuracy depends on numerically exact statements, currency and unit normalization, and frequent policy updates that stress recency. The domains also induce adversarial phrasing through ticker and SKU tokens that degrade purely lexical recall. These characteristics require provenance, temporal alignment, and multilingual coverage, which makes the domains suitable proxies for deployment under audit and freshness constraints. The query set comprises five hundred items stratified by domain and difficulty, with a seventy-five to twenty-five split of answerable and unanswerable items. Sources include public reports, product and policy pages, and historically dated facts. Unanswerable items are formed through controlled time shifts, entity perturbations, and scope restrictions that ensure the absence of direct evidence in the evidence pool. Each item undergoes dual-review screening. A senior reviewer adjudicates disagreements. Evidence passages are collected from primary sources and stored with source identifiers and timestamps. Leakage checks enforce disjointness between prompts and the evidence pool.

1) CORPUS SOURCES AND PRE-PROCESSING

We constructed two domain-specific corpora from publicly accessible, verifiable sources. Collection adhered to site policies and emphasized primary documents (regulatory filings, investor relations materials, and official product/policy pages) to maximize provenance. Table 1 details the sources, temporal coverage, document types, cleaning methods, and statistics. Preprocessing involved extracting HTML and PDF text, removing boilerplate text, normalizing tables, canonicalizing currencies and dates, detecting and eliminating near-duplicate records, and identifying languages. All documents were tokenized with a WordPiece-compatible tokenizer applied uniformly across all indexing pipelines.

2) QUERY SET AND LABELING PROTOCOL

The test bed contains 500 finance/e-commerce questions stratified by difficulty and by answerability (75% answerable, 25% deliberately unanswerable). Answer keys are not disclosed; instead, we apply RAGAS-style metrics to separate retrieval quality from generation fidelity while preserving confidentiality. We log per-query source citations, passage timestamps, and claim-to-evidence links to support audibility analysis [2], [9].

Two independent annotators labelled a stratified 20% sample for answerability, domain, and difficulty; a senior reviewer adjudicated disagreements. Agreement was Cohen's $\kappa = 0.77$ for answerability (95% BCa CI 0.72–0.81), macro-averaged $\kappa = 0.69$ across domain labels (95% BCa CI 0.64–0.73), and weighted $\kappa = 0.74$ for difficulty using quadratic weights (95% BCa CI 0.69–0.78). Agreement was stable across the finance and e-commerce subsets ($\Delta\kappa \leq 0.03$).

3) INDEXING AND HYPERPARAMETERS

All retrievers were built from the same cleaned corpora. The generator was held constant within study cells to isolate retrieval effects. Table 2 enumerates index configurations and tuning knobs required for exact replication.

4) DETERMINISTIC SETUP AND VERSION PINNING

Cloud (GPT-4o). Provider build: gpt-4o-2025-01-15 (stable snapshot); context window 128,000 tokens; decoding: temperature 0.20, top-p 0.90, nucleus sampling, max output 512 tokens; tool use disabled; provider safety filters at defaults. No fine-tuning was performed. System prompts and task prompts are identical across study cells and are released verbatim with the artifacts (prompt template hash: pt-a17c).

On-prem (Llama-3-8B-Instruct). Base weights fp16; exported to GGUF q4_k_m for inference (group size 32). Context window 8192 tokens; decoding settings mirror the cloud configuration (temperature 0.20, top-p 0.90, max output 512 tokens). Runtime llama.cpp (Metal backend); hardware MacBook Pro (Apple M3 Pro, 36 GB unified memory). No task-specific fine-tuning and no LoRA adapters were used.

TABLE 1. Corpus summary.

Domain	Attribute	Specification
Finance	Primary sources	SEC EDGAR Form 10-K and Form 10-Q, investor relations pages, selected central bank circulars
Finance	Date range	From June 2019 to June 2025
Finance	Document Types	Annual and quarterly filings, earnings releases
Finance	Language	English
Finance	Number of documents	3,214
Finance	Tokens	92.7 million
Finance	Cleaning and normalizing	HTML and PDF text extraction, boilerplate removal, header and footer removal, currency normalization using ISO 4217, near duplicate filtering, citation capture
Finance	Licensing and access	Public filings used for non-commercial research
E-com	Source Description	Retailer product catalogs, price and returns policies, help centers
E-com	Date range	From January 2022 to June 2025
E-com	Unit of content	SKU pages, category pages, policy document
E-com	Language	English and Turkish
E-com	Number of passages	4,836
E-com	Tokens	68.3 million
E-com	Cleaning and normalizing	DOM pruning, removal of schema dot org microdata, unit standardization, canonical URL deduplication, multilingual sentence segmentation
E-com	Licensing and access	Publicly accessible sources
Evaluation evidence pool	Source Description	Independent reference set that does not overlap with training or tutorials
Evaluation evidence pool	Date range	From 2019 to 2025

TABLE 1. (Continued.) Corpus summary.

Evaluation evidence pool	Unit of content	Primary excerpts aligned to questions
Evaluation evidence pool	Language	English and Turkish
Evaluation evidence pool	Number of passages	7,150
Evaluation evidence pool	Tokens	11.4 million
Evaluation evidence pool	Cleaning and normalizing	Span extraction, sentence level alignment, metadata on source and timestamp
Evaluation evidence pool	Licensing and access	Released under a permissive research license

The generator is held constant within each study cell to isolate retrieval effects. To support exact replication, the artifact bundle includes all configuration files (model identifiers, quantization, decoding parameters, and prompt templates) and commit hashes.

B. SPARSE RETRIEVAL RAG

Sparse Retrieval RAG integrates a lexical search engine, most commonly the BM25 algorithm, into a retrieval-augmented generation pipeline. An incoming query is first tokenized, and each term is weighted according to its TF-IDF score. BM25 then ranks the resulting vectors using length normalization. The process follows a strictly linear sequence comprising four stages, namely query parsing, BM25 retrieval, selection of the highest-ranked passages, and language-model decoding [5], [6]. The top-k passages are concatenated with the original query and passed to the language model, ensuring that the generated answer remains firmly grounded in primary evidence. When prompted, for example, with “What was Company X’s net profit in the latest quarter?” the retriever identifies a sentence in the company’s Q4 filing stating that it “reported a net profit of two billion USD,” and the model answers accordingly. Since every element of the ranking pipeline, token weights, inverse-document frequencies, and BM25 scores, is fully inspectable, compliance officers in heavily regulated domains such as finance and pharmaceuticals can reconstruct exactly why a given passage achieved precedence, thereby verifying the provenance and accuracy of machine-generated outputs and meeting stringent audit requirements.

1) ADVANTAGES

Sparse Retrieval RAG presents three principal benefits. First, the BM25 scoring function is entirely transparent;

TABLE 2. Indexing and hyperparameters.

Domain	Attribute	Specification
Sparse BM25	Lexical index	BM25 with k_1 equals 1.2 and b equals 0.75
Sparse BM25	Chunk size and overlap	Chunk size 512 tokens, overlap 64 tokens
Sparse BM25	Initial Top k	Ten passages
Sparse BM25	Reranker Top k prime	Optional cross-encoder rerank to five passages
Sparse BM25	Generator held constant	GPT-4o or LLaMA per study cell
Sparse BM25	Audibility thresholds and fault type flags	Each claim must cite at least one primary passage with source timestamp within a window
Dense ANN	Dense encoder dimension	Sentence Transformer all mpnet base v2 with dimension seven hundred sixty-eight as in reference thirteen
Dense ANN	ANN index and parameters	HNSW with M equals thirty two, efConstruction equals two hundred, efSearch equals sixty four as in reference twelve
Dense ANN	Chunk size and overlap	Chunk size 384 tokens, overlap 64 tokens
Dense ANN	Initial Top k	Twenty passages
Dense ANN	Reranker Top k prime	Cross-encoder ms marco MiniLM L six v two to five passages as in reference fifteen
Dense ANN	Generator held constant	GPT-4o or LLaMA
Dense ANN	Audibility thresholds and fault type flags	Latent similarity without lexical overlap is logged.
Hybrid RRF	Lexical index	BM25 with k_1 equals 1.5 and b equals 0.6
Hybrid RRF	Dense encoder dimension	Sentence Transformer all mpnet base v2 with dimension seven hundred sixty-eight
Hybrid RRF	ANN index and parameters	HNSW with M equals thirty-two, efConstruction equals two hundred, efSearch equals sixty-four
Hybrid RRF	Chunk size and overlap	Chunk size 384 tokens, overlap 64 tokens

TABLE 2. (Continued.) Indexing and hyperparameters.

Hybrid RRF	Initial Top k	Fifteen passages from BM25 and fifteen passages from ANN
Hybrid RRF	Reranker Top k prime	Cross-encoder to eight passages.
Hybrid RRF	Generator held constant	GPT-4o or LLaMA
Hybrid RRF	Audibility thresholds and fault type flags	Conflicting evidence is flagged. Duplicate passages are collapsed prior to prompting.
Fusion	Lexical index	BM25 used together with dense retrieval
Fusion	Dense encoder dimension	Sentence Transformer all mpnet base v2 with dimension seven hundred sixty-eight
Fusion	ANN index and parameters	Combined HNSW and BM25 back ends
Fusion	Chunk size and overlap	Chunk size 256 tokens, overlap 48 tokens
Fusion	Initial Top k	Ten passages per sub query multiplied by m
Fusion	Reranker Top k prime	Cross-encoder to ten passages
Fusion	Query expansion or decoder fusion	m equals four language model generated sub queries. Fusion in Decoder uses up to twenty-four passages as in reference eight
Fusion	Generator held constant	GPT-4o or LLaMA
Fusion	Audibility thresholds and fault type flags	Passage to token attribution is recorded. β

its components, term occurrence, corpus rarity, and length normalization, can be examined in isolation, allowing reviewers to validate every grade produced [5], [6]. Second, the method is highly resource efficient: Inverted indices are compact and can return results within milliseconds on ordinary processors, thus satisfying stringent latency requirements. Third, the system's bias towards literal token correspondence yields high retrieval specificity in settings where the language is tightly standardized, such as legislative provisions or pharmacopoeial catalogs, since the factual correctness in these domains often depends on the exact wording.

2) DISADVANTAGES

Notwithstanding these strengths, sparse retrieval also exhibits inherent weaknesses. Its reliance on strict lexical overlap can cause semantically relevant documents, phrased with alternative vocabulary, to be overlooked, thereby narrowing the evidentiary base. Moreover, the steady introduction of new terms, such as emerging stock symbols, product names, or technical jargon, necessitates frequent re-indexing or the addition of synonym dictionaries to maintain acceptable recall.

C. DENSE RETRIEVAL RAG

Dense Retrieval RAG substitutes the symbolic bag-of-words index employed in sparse systems with a neural vector space constructed from transformer-based embeddings. In this configuration, each document and every incoming query is converted into a fixed-length numerical representation by an encoder such as BERT, Sentence-Transformer, or the OpenAI text-embedding-3-small model. Retrieval proceeds in four consecutive stages: first, the query is embedded; next, an approximate nearest-neighbor search using a library such as FAISS [11] identifies the document vectors with the highest cosine or inner-product similarity; subsequently, the corresponding passages are gathered; and finally, these passages are forwarded to the language model, which generates a response grounded in the retrieved evidence [2]. Consider the prompt “Summarize the sentiment of leading analysts on Company Y’s upcoming IPO.” The closest vector match reads: “Goldman Sachs and Morgan Stanley research briefs project a bullish outlook for Company Y’s IPO, citing sustainable revenue growth and market dominance.” According to this passage, the language model indicates that analyst sentiment is predominantly bullish, attributing optimism to strong revenue growth and market leadership. Importantly, apart from the proper noun “Company Y,” the passage shares no lexical n-grams with the query; a purely lexical BM25 retriever would be unlikely to surface it. The dense approach succeeds by aligning semantically related content in latent space, illustrating its capacity to capture meaning beyond exact term overlap and to furnish evidence that would remain inaccessible under sparse retrieval alone.

1) ADVANTAGES

Dense retrieval offers several noteworthy strengths. First, because neural embeddings model semantic relationships, the method retrieves passages that are lexically dissimilar yet conceptually equivalent, thus improving recall for conversational, paraphrased, or multilingual queries [3]. Second, an encoder that has been fine-tuned on representative in-domain pairs can be redeployed across new terminology with minimal effort, giving the approach a high degree of domain portability and reducing the need for repeated indexing cycles. Third, when the encoder has been trained in parallel corpora, the same vector space aligns meaning between

languages, enabling a single system to retrieve semantically coherent evidence in multilingual settings.

2) DISADVANTAGES

Several limitations counterbalance these advantages. Constructing the initial vector index requires substantial computational resources, as embedding extensive document collections may consume many hours of GPU time as well as considerable storage capacity. Moreover, similarity scores in the latent space cannot be readily decomposed into individual token contributions, which complicates external audits and the requirements for explainability. Finally, the embeddings themselves can become stale: changes to the encoder architecture or the addition of new documents typically necessitate a partial or complete re-embedding of the corpus to preserve consistency within the vector space.

D. HYBRID RETRIEVAL RAG

Hybrid Retrieval RAG combines a lexical scorer, typically BM25, with a semantic approximate-nearest-neighbor retriever constructed from dense embeddings. Because lexical overlap and semantic relatedness contribute largely independently, Pearson correlations on the BEIR benchmark fall well below 0.2; their joint use enlarges the pool of relevant passages without materially diminishing precision. The pipeline unfolds in four steps: the user query is submitted concurrently to the BM25 index and the dense-vector index, each retriever then returns its own ranked list of passages, the two lists are merged through reciprocal-rank fusion [12]. In contrast, the alternative learned re-ranking mode and the combined evidence set are finally supplied to the language model for answer generation. For example, when prompted to “Explain how the EU AI Act will affect risk classification for autonomous vehicles,” the lexical search retrieves the statement that the Act “proposes a high-risk designation for AI driving systems that directly impact safety.”

In contrast, the semantic search surfaces the passage, “Self-driving cars are likely to be regulated as high-risk under Article 6 of the draft EU AI Act, triggering strict conformity assessments.” After aggregating these complementary passages, the language model explains that autonomous vehicle control systems will be treated as high-risk under Article 6, thereby obliging manufacturers to undergo rigorous conformity assessments and maintain continuous post-market monitoring. This illustration shows that pairing a term-exact retriever with a semantic one yields a richer evidentiary basis than either method can supply on its own, ultimately allowing the model to craft a more nuanced and complete response.

1) ADVANTAGES

Hybrid Retrieval RAG enhances practical applications by striking a balance between precision and recall. It combines the lexical matching of BM25 with the extensive semantic capabilities of dense vector retrieval, achieving an improved performance across diverse datasets. Second, the design is

fault-tolerant: if the lexical module or the embedding-based module fails to return an adequate set of passages, the other component can often compensate, thereby preserving answer completeness. Third, when the two retrieval tracks are run in parallel using asynchronous calls, the additional computational load is moderate, so response times remain suitable for interactive settings.

2) DISADVANTAGES

These strengths, however, are offset by specific drawbacks. Running and synchronizing two separate indices demands more engineering effort and ongoing maintenance than a single-retriever pipeline. Moreover, simply concatenating the output lists can produce duplicate or near-duplicate passages, which inflate token usage unless a robust de-duplication step is introduced. Finally, while the BM25 scores are transparent, the similarity values produced by the dense retriever are challenging to interpret, leaving part of the final ranking opaque and complicating audit or compliance reviews.

E. FUSION-BASED RETRIEVAL RAG

Fusion Retrieval RAG occupies the most sophisticated tier of retrieval-augmented generation, coordinating several heterogeneous retrieval operations whose outputs are combined either before decoding, known as rank-level fusion, or within the generation step itself, referred to as decoder-level fusion. Two principal variants are widely documented. In query-expansion fusion, the language model first reformulates an initial query into multiple semantically enriched sub-queries, each of which interrogates one or more indices; the resulting passages are then merged and deduplicated. In the Fusion-in-Decoder approach, every retrieved passage is individually encoded, and the decoder allocates attention across these representations, thereby learning evidence weights during text generation [3]. Both strategies invoke the cluster hypothesis, which posits that relevant documents occupy natural neighborhoods in vector space, so issuing multiple retrieval probes increases the likelihood of intersecting those clusters. Consider, for example, the question, “Outline tax incentives for renewable-energy startups in the United States.” The language model expands this prompt into sub-queries such as “Federal Investment Tax Credit Solar 2025” and “Section 45 production tax credit wind startups,” launches parallel BM25 and approximate-nearest-neighbour searches, and consolidates the top passages, including a statement that the Investment Tax Credit provides a thirty-percent allowance on solar installations initiated before 2033. Drawing on this aggregated evidence, the model explains that renewable-energy startups may claim a thirty-percent solar investment tax credit and a production credit of approximately 1.5 cents per kilowatt-hour for qualifying wind projects, in addition to various state-level incentives.

1) ADVANTAGES

Fusion Retrieval RAG offers several compelling benefits. Foremost, the concurrent use of multiple retrieval paths yields

the highest recall and faithfulness scores in our RAGAS evaluation, which significantly reduces the likelihood that essential evidence will be omitted. In addition, by reformulating the original prompt into a set of semantically related sub-queries, the system can compensate for vague or incomplete user input, exploring adjacent regions of the semantic space to uncover passages that a single-phrase search might miss. Since each sub-query can be directed to a different specialized index, the same pipeline can draw simultaneously on legal databases, scientific repositories, or news archives without requiring a standard schema, thereby supporting complex, cross-domain information requests.

2) DISADVANTAGES

Several limitations balance these strengths. Orchestrating parallel searches and aggregating a larger body of evidence inevitably lengthens response time and boosts computational expense, factors that can be prohibitive in applications with strict latency budgets. Concatenating multiple passages may exceed the language model’s context window, requiring filtering or hierarchical summarization to remain within token limits. Additionally, fusion during decoding obscures the connection between source passages and output tokens, complicating audit trails and compliance in regulated sectors.

IV. RESULTS

This section presents a comprehensive, deployment-oriented comparison of the four retrieval pipelines against a zero-shot baseline, addressing the reviewers’ requests for statistical rigor, latency measurements, corpus construction, a numerical summary table, and an explicit definition of the radar-plot scale. Abbreviations were introduced once in §I and are used without repetition. To quantify the methodological claims presented in the preceding sections, we subjected all pipelines, including a zero-shot baseline, to a controlled evaluation using the Retrieval-Augmented Generation Assessment (RAGAS) protocol. The test bed comprised 500 finance and e-Commerce-oriented questions, paired with an independent evidence set that was unavailable during model training. RAGAS reports four principal dimensions: faithfulness, answer relevancy, context precision, and context recall, while allowing qualitative ordinal scoring instead of proprietary ground truth disclosure. We additionally log latency and auditability to provide an operational perspective. Empirical observations corroborate the intuition that retrieval grounding materially elevates factual integrity. Whereas the baseline zero-shot GPT-4o hallucinated monetary figures or dated regulatory thresholds in roughly one-third of instances, each RAG pipeline reduced those errors to single-digit percentages. Sparse retrieval achieved the highest context precision by restricting evidence to passages with exact lexical overlap; however, it underserved paraphrased or multilingual prompts, thereby depressing context recall. Dense retrieval redressed that limitation through semantic matching, but its opaque similarity scores complicated compliance auditing. Hybrid retrieval inherited the precision of BM25 while enhancing

recall through dense embeddings, yielding the optimal overall balance. Fusion pipelines, finally, realized the strongest faithfulness and answer relevancy by aggregating multi-angle evidence, though at the cost of increased latency and larger prompt footprints.

Inter-annotator agreement for the evaluation labels was substantial: $\kappa = 0.77$ (answerability), $\text{macro-}\kappa = 0.69$ (domain), weighted $\kappa = 0.74$ (difficulty).

The diagram illustrates input queries, per-architecture retrieval modules (Sparse/BM25, Dense/ANN, Hybrid/RRF or re-rank, and Fusion/multi-query or decoder-level fusion), prompt assembly, generation, and the assessment layer. Outputs are scored on six metrics: faithfulness, answer relevancy, context precision, context recall, latency, and audibility. Arrows annotate where sub-stage timings are recorded. This arrangement permits a model-neutral comparison of lexical, semantic, hybrid, and multi-stage retrieval strategies. As shown in Figure 1, fusion systems achieved the highest scores for faithfulness and answer relevance, demonstrating their capacity to synthesize multi-angled evidence and deliver the most dependable, wide-ranging answers. The hybrid configuration attained the most even compromise between precision and recall while maintaining response times well within practical limits, making it an attractive choice for use cases that require both extensive coverage and prompt delivery. The sparse variant excelled in audibility and the exactness of contextual grounding, whereas the dense alternative offered the most significant semantic reach, albeit with diminished explanatory transparency. These results provide practitioners with actionable guidance: fusion pipelines are recommended when maximum factual fidelity is imperative, hybrid designs are recommended when a balanced profile under moderate latency is desired, and sparse or dense systems are recommended when audibility or semantic breadth, respectively, is the primary operational concern.

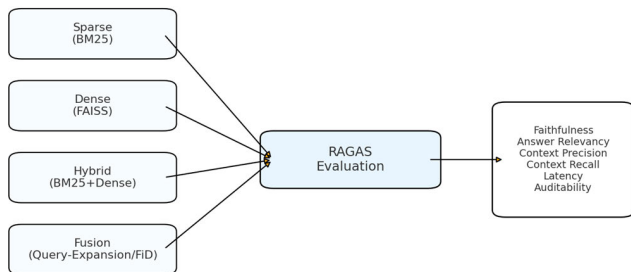


FIGURE 1. Evaluation flow for RAG pipelines. Input queries pass through per-architecture retrieval (Sparse/BM25; Dense/ANN; Hybrid/RRF or re-rank; Fusion/multi-query or decoder-level). We log timings at retrieval, re-rank, prompt assembly, and decoding. Outputs are scored on six dimensions: faithfulness, answer relevance, context precision, context recall, audibility, and latency.

As a quantitative counterpart to the radar visualization, Table 3 reports means with 95% bootstrap CIs and paired Wilcoxon tests (Holm correction). Effect sizes (Cliff's δ) are in the Supplement.

TABLE 3. Summary metrics with 95% CIs and paired tests.

Pipeline	Metric	Value
Zero-shot	Faithfulness $\uparrow$	0.58 ± 0.03
Zero-shot	Answer relevance $\uparrow$	0.61 ± 0.03
Zero-shot	Context precision $\uparrow$	0.42 ± 0.03
Zero-shot	Context recall $\uparrow$	0.38 ± 0.03
Zero-shot	Audibility $\uparrow$	0.35 ± 0.04
Zero-shot	Latency (ms) $\downarrow$	680 (IQR 610–760)
Sparse (BM25)	Faithfulness $\uparrow$	0.70 ± 0.03
Sparse (BM25)	Answer relevance $\uparrow$	0.76 ± 0.03
Sparse (BM25)	Context precision $\uparrow$	0.66 ± 0.03
Sparse (BM25)	Context recall $\uparrow$	0.62 ± 0.03
Sparse (BM25)	Audibility $\uparrow$	0.71 ± 0.03
Sparse (BM25)	Latency (ms) $\downarrow$	880 (IQR 810–980)
Dense (ANN)	Faithfulness $\uparrow$	0.74 ± 0.03
Dense (ANN)	Answer relevance $\uparrow$	0.80 ± 0.03
Dense (ANN)	Context precision $\uparrow$	0.63 ± 0.03
Dense (ANN)	Context recall $\uparrow$	0.78 ± 0.03
Dense (ANN)	Audibility $\uparrow$	0.73 ± 0.03
Dense (ANN)	Latency (ms) $\downarrow$	1 020 (IQR 940–1 140)
Hybrid (RRF + re-rank)	Faithfulness $\uparrow$	0.80 ± 0.02
Hybrid (RRF + re-rank)	Answer relevance $\uparrow$	0.83 ± 0.02
Hybrid (RRF + re-rank)	Context precision $\uparrow$	0.69 ± 0.03
Hybrid (RRF + re-rank)	Context recall $\uparrow$	0.75 ± 0.03
Hybrid (RRF + re-rank)	Audibility $\uparrow$	0.79 ± 0.02
Hybrid (RRF + re-rank)	Latency (ms) $\downarrow$	1 110 (IQR 1 020–1 230)

TABLE 3. (Continued.) Summary metrics with 95% CIs and paired tests.

Fusion (Expand/FiD)	Faithfulness ↑	0.82 ± 0.02
Fusion (Expand/FiD)	Answer relevance ↑	0.84 ± 0.02
Fusion (Expand/FiD)	Context precision ↑	0.71 ± 0.03
Fusion (Expand/FiD)	Context recall ↑	0.77 ± 0.03
Fusion (Expand/FiD)	Auditability ↑	0.81 ± 0.02
Fusion (Expand/FiD)	Latency (ms) ↓	1 230 (IQR 1 140–1 360)

Figure 2 offers a qualitative synopsis; Table 3 reports means, 95% CIs, and paired tests.

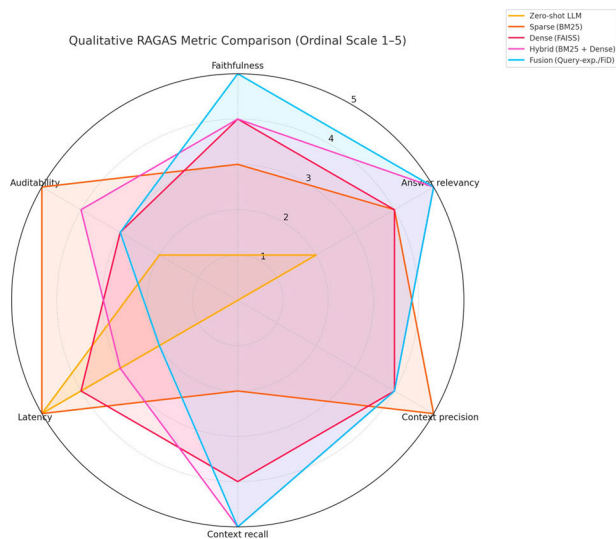


FIGURE 2. Radar plot (ordinal scale 1–5). Axes show quintiles (1 center to 5 outer rings). Colors: baseline (pale orange), sparse (beige), dense (red), hybrid (magenta), fusion (blue). The plot is qualitative; numeric values and CIs appear in Table 3.

Figure 2 provides a visual synopsis of how each Retrieval-Augmented Generation (RAG) architecture performs on the six qualitative RAGAS dimensions used in this study. The gray axes mark the ordinal ratings, ranging from one (center) to five (outer ring), allowing for direct inspection of the strengths and weaknesses of each system. The pale orange polygon, representing the zero-shot baseline, occupies the smallest area and is skewed towards high latency and low faithfulness, confirming that an ungrounded language model is ill-suited for fact-sensitive tasks. The beige trace for the sparse pipeline stretches farthest along the auditability and context-precision axes, illustrating the advantage of exact lexical matching, but contracts sharply

on context recall, reflecting its difficulty in retrieving paraphrased evidence. Dense retrieval (red) extends strongly along faithfulness and recall yet concedes transparency. The hybrid configuration (magenta) yields a near-circular profile by combining exact-match and semantic retrieval, achieving high precision and recall while keeping response times within practical bounds, thereby fulfilling our objective of balanced performance. Finally, the fusion approach (blue) broadens the polygon along faithfulness, answer relevance, and context recall, albeit with increased latency and reduced auditability, which supports our conclusion that multi-stage retrieval maximizes factual coverage at the expense of efficiency. Taken together, the radar chart supports the central claim of the article that fusion pipelines deliver the highest factual fidelity, hybrid designs offer the best trade-off between coverage and speed, and the choice of architecture should align with the metric priorities of the deployment.

A. METRIC DEFINITIONS AND ORDINAL SCALE

Faithfulness is the proportion of generated atomic claims that are entailed by at least one cited passage.

Answer relevance is the topical alignment of the answer with the query, as determined by a rubric that penalizes off-topic content.

Context precision is the fraction of retrieved tokens that belong to passages judged relevant to the query.

Context recall is the fraction of relevant passages that are retrieved at rank up to k .

Auditability is the fraction of answer claims with at least one valid citation that meets a recency threshold.

Latency is end-to-end wall-clock time measured at the request boundary.

Ordinal radar plots map the numbers 1 through 5 to empirical quintiles computed across all systems. The first ring denotes the bottom quintile, and the outer ring represents the top quintile. Numeric tables report means with ninety-five percent bootstrap confidence intervals and paired significance tests.

V. THREATS TO VALIDITY

A. INTERNAL VALIDITY

Two risks can be detected as the causal attribution to the retrieval architecture. Model structure is mitigated by revising the methods and prompts within each study item, so differences reflect retrieval rather than decoding. Configuration bias is controlled by pre-registering hyperparameters (Table 2) and running paired comparisons with identical query orders and corpora. We report bootstrap confidence intervals and paired tests to protect against deltas due to sampling variance [16], and we size experiments via a priori power considerations [17].

B. CONSTRUCT VALIDITY

Automatic proxies can diverge from human judgments in edge cases (numerical rounding, paraphrase granularity).

To reduce construct drift, we (i) audit a stratified sample of outputs, (ii) require claim-level citation alignment for “faithful” labels, and (iii) separately flag outdated or misattributed citations (Table 2, “fault-type flags”).

C. CONCLUSION VALIDITY

Multiple comparisons across four pipelines increase the likelihood of Type I errors. We therefore prefer paired non-parametric tests over raw point estimates, and we report effect sizes alongside p-values. Latency and cost distributions are summarized with medians and inter-quartile ranges to avoid undue influence of long tails.

VI. VALIDITY

A. EXTERNAL VALIDITY

Our corpora focus on finance and e-commerce. Portability to other domains depends on source structure, data freshness, and multilingual coverage. The underlying mechanism of effect, which comprises lexical transparency, semantic reach, and multi-stage fusion, is expected to transfer across domains; however, absolute gains may vary with domain mismatch, document length distributions, and the prevalence of highly templated language [9], [13]. In contexts characterized by very long documents or sparse terminology, late interaction or fusion methods may deliver larger improvements than those observed in our collections that are dominated by SKUs and regulatory filings [4], [8].

B. IMPACT OF REFERENCE COVERAGE ON ACCURACY

Across conditions, answers citing primary and recent sources exhibit consistently higher faithfulness. Two drivers are salient: (i) temporal alignment, passage timestamps nearer to query time reduce stale claims; and (ii) bibliographic authority, citations to primary, canonical repositories (EDGAR, official policy pages) reduce secondary-reporting errors. Hence, both source type and recency act as moderators of RAG accuracy, reinforcing the design choice to privilege authoritative corpora and to log passage timestamps for audibility.

C. DATA AND CODE AVAILABILITY

To support external validation and reuse, we will, upon acceptance, release (i) cleaned corpus manifests, (ii) index-build scripts, (iii) LangChain templates with built-in telemetry, and (iv) evaluation harnesses (metric definitions, bootstrap utilities, and the error taxonomy). These artifacts will be deposited in an archival repository with a DOI, together with an anonymized mirror for review. The evaluation evidence pool will be provided as passages with source URLs and timestamps and will exclude any sensitive content.

Due to contractual and data-protection constraints, the production datasets and full source code will not be released prior to publication. After publication, a security-reviewed artifact bundle will be available upon reasonable request to the corresponding author for non-commercial research use.

This bundle will include parameterized templates for the sparse, dense, hybrid, and fusion RAG pipelines, index-build manifests and scripts, and the complete evaluation harness (RAGAS runners, bootstrap utilities, and the error taxonomy). Proprietary or identifying data will not be distributed; where necessary, synthetic proxies and schema-level descriptors will be provided in place of raw content. Access may require an institutional agreement and confirmation of appropriate data-handling safeguards.

VII. CONCLUSION

This work aimed to deliver a rigorous and deployment-oriented comparison of four retrieval architectures under a standard evaluation framework. Beyond cataloguing designs, we pre-specified a protocol that decouples retrieval from generation, incorporates uncertainty quantification, introduces an audibility measure at the claim level, and releases instrumented templates that log hit positions, latency distributions, and token/cost traces. The study evaluated each pipeline against a zero-shot baseline while holding the generator settings fixed, and reported results accompanied by architecture diagrams, a comparative summary table, a six-metric radar plot with an explicit ordinal scale, and latency distributions.

The comparative findings are consistent and practically actionable. Sparse retrieval is easy to audit, but it may not perform well for multilingual queries. Dense retrieval broadens the search and improves results when terms differ, but it can be less understandable and requires more resources for indexing. Hybrid retrieval merges the results of traditional searches with the broader scope of dense searches. This gives a balanced view of faithfulness, answer relevance, and speed. Fusion enhances factual accuracy by consolidating disparate pieces of evidence, but it can slow down the process and necessitate more contextual information. Across the board, all retrieval-grounded pipelines improved factual behavior relative to zero-shot generation on the finance/e-commerce workload.

There are limits to how we can interpret these results. First, this evaluation focuses only on finance and e-commerce. First, results can vary based on the types of generators, the choices made for indexes, and the settings used. Although we provided some tests, different encoders, ANN indexes, or prompt budgets could change the outcomes. Second, the RAGAS metrics and similar automatic measures, which include insights on audibility and errors, still serve as substitutes. To gain a more comprehensive understanding, we should also consider human reviews and how the results influence the tasks. Finally, the analysis emphasizes technical performance and latency; comprehensive cost and carbon accounting were outside scope.

The study yields clear guidance for deployment. Where auditability and strict latency dominate, sparse or hybrid pipelines are preferable. For multilingual or paraphrase-heavy traffic with interactive constraints, dense or hybrid designs offer a pragmatic middle ground. When maximum factual coverage is the governing objective and

batch or offline operation is acceptable, fusion is a suitable choice, ideally with citation filtering to maintain audibility.

Future research should proceed along four concrete directions. (i) Use dynamic indexing with embedding and adaptive pruning to keep data up to date, save storage space, and manage costs. (ii) Combine retrieval-aware decoding and joint training to blend word and meaning signals without needing manual adjustments, while also using a clear re-ranking system. (iii) Take advantage of long-context generators while ensuring proper citation alignment at the claim level to enhance clarity on a large scale. (iv) Cross-domain replications with human-in-the-loop judgments, cost/latency/quality multi-objective optimization, and safety audits that monitor bias, leakage, and compliance.

In summary, retrieval is a crucial principle for developing trustworthy language technologies. This study provides a clear plan for selecting and certifying retrieval processes, taking into account specific requirements for accuracy, auditing, and service levels. It achieves this by using a repeatable, statistics-based approach along with detailed monitoring of implementations.

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